

Muhammad Usman Butt

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer specializing in real-time multiplayer and VR experiences in Unity (Netcode, Photon PUN 2, Meta Quest), with additional experience in LLM-driven systems (RAG pipelines, agentic game AI) and full-stack web and mobile (React, Node.js, MongoDB, Redis). Placed 5th out of 200 teams at the Google Antigravity Mobile Game Challenge 2026 and won 1st place at Softcom 2026. Led a game development department and shipped cross-platform titles from concept to deployment.

TECHNICAL SKILLS

Languages: C#, Python, JavaScript, TypeScript, HTML, CSS

Game Engine & Multiplayer: Unity (2D/3D), ScriptableObjects, Unity Physics, Animation Systems, State Machines, Object Pooling, Netcode for GameObjects, Photon PUN 2, Unity Relay, Unity Lobby, Host Migration, Reconnection Handling

VR & Immersive Systems: Meta Quest Development, XR Interaction Toolkit, VR Interaction Framework, Final IK, VR Locomotion, Multiplayer VR, Gaussian Splatting

LLMs & RAG: OpenAI API, Groq API, Gemini, NVIDIA NIM, Retrieval-Augmented Generation, Prompt Engineering, Embeddings, Semantic Search, LLM-Driven Game Agents, Procedural Generation

Frontend & Mobile: React, React Native, Android, WebGL, Cross-Platform Build Pipelines

Backend & Data: Node.js, Express.js, REST APIs, OAuth, BullMQ, MongoDB, Redis, Firebase (Auth, Realtime Database), Cloud Save Systems, Unity IAP, Google AdMob

Tools & Practices: Git, GitHub, CI/CD, Agile/Scrum, Quality Assurance, Team Leadership

PROFESSIONAL EXPERIENCE

AI & Full-Stack Engineer Intern | Termnl Tech

Jun 2026 – Sep 2026

- Built a full-stack RAG platform (React, Node.js/Express, MongoDB, Redis, BullMQ) enabling natural-language queries over connected data sources and uploaded documents, with scoped OAuth integrations and background ingestion jobs.
- Developed features for a React Native Android app, including an AI image generation feature driven by user prompts and in-app selection.
- Worked with the team on quality assurance, testing, debugging, and resolving frontend and feature issues.

Head of Game Development | Axiolink Game Studio

Apr 2026 – Jun 2026

- Led the game development department, directing the full production pipeline from concept to implementation for Unity-based, systems-driven games.
- Set development strategy and coordinated a cross-functional team while remaining hands-on in C# engineering.
- Drove integration of agentic AI into gameplay systems as a core studio direction.

Unity Developer | AstroApe Studios

Sep 2025 – Apr 2026

- Engineered real-time multiplayer features enabling seamless player interactions across game builds.
- Optimized application performance and scalability, ensuring a smooth user experience across platforms.
- Collaborated with a creative team in a hybrid environment, merging artistic vision with technical execution.

Unity Developer | Fiverr (Freelance)

Mar 2023 – Present

- Designed and developed immersive 3D VR experiences using Unity, VR Interaction Framework, Final IK, and Photon PUN 2 for multiplayer environments.
- Solved complex technical challenges to deliver high-quality VR builds aligned with client goals and player expectations.
- Delivered 3+ years of continuous freelance client work across VR and multiplayer game projects.

TECHNICAL PROJECTS

RAG Platform | Full-Stack & AI (Termnl Tech)

Stack: React, Node.js/Express, MongoDB, Redis, BullMQ, OpenAI, Groq

Designed a multi-model retrieval pipeline: OpenAI for answer generation, Groq for low-latency reranking and intent classification. Added Redis caching to cut repeat-query cost and improve response speed.

THRESHOLD | Agentic AI Top-Down Shooter

Stack: Unity, C#, Gemini/Groq/NVIDIA NIM | github.com/usmanbutt-dev/ThresholdGame

Built a 5-agent LLM pipeline driving procedural dungeon generation, adaptive difficulty, and real-time NPC combat decisions (including NPC defection); placed 5th of 200 teams at Google Antigravity Mobile Game Challenge 2026.

ShuffleMania | Mobile Multiplayer Game Collection (Client Project)

Stack: Unity, Netcode for GameObjects, Firebase, Unity IAP, AdMob

Shipped a 5-game mobile collection (3 solitaire variants, real-time multiplayer Ludo and Rummikub) with cloud saves, public/private lobbies, host migration handling, and full monetization (IAP + rewarded ads) across Android and iOS.

A House on Larcombe Road | Co-op Horror, Unity (Client Project)

Stack: Unity, C#, Multiplayer Networking

Built a 2-player psychological co-op horror game with asymmetric twin characters, dual-input puzzle mechanics requiring synchronized actions, and a dynamic sanity system driving vision distortion, stealth detection, and light/sound-based enemy AI.

2D Action Platformer

Stack: Unity, C#

Built a geometry-dash-style 2D action platformer; won 1st place at Softcom 2026 (GIKI).

Unity Modular Inventory System | Open-Source Tool

Stack: Unity, C#, ScriptableObjects | github.com/usmanbutt-dev/UnityModularInventorySystem

Designed a reusable, extensible inventory framework using ScriptableObject-based item data and clean data/logic/UI separation for drop-in use across RPG, survival, and crafting games.

For more projects and source code, check out: usmanbutt.dev

EDUCATION

BS Computer Science | UMT, Lahore | CGPA 3.50/4.00

Aug 2022 – Aug 2026

CERTIFICATIONS

- Game Design and Development 1: 2D Shooter – Michigan State University (Coursera)